Revisiting the Zero Lower Bond Issue with A Conceptual Trinity of Macroeconomic Analysis

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Statement of Originality

I hereby declare that this submission is my own work and to the best of my knowledge it contains no material which has been accepted for the award of any other degree or diploma at the University of Sydney or at any other educational institution, except where due acknowledgement is made in this thesis.

Any contributions made to the research by others with I have had the benefit of working with at the University of Sydney is explicitly acknowledged. I also declare that the intellectual content of this study is the product of my own work and research, except to the extent that assistance from others in the project's conception and design is acknowledged.

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Revisiting the Zero Lower Bound Issue with A Conceptual Trinity of Macroeconomic Analysis

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This paper revisits the zero lower bound (ZLB) issue in a two-asset Heterogeneous Agent New Keynesian (HANK) model with real wage rigidity and proposes high-level characteristic framework for shock and policy analysis – the trinity of resource flows, economic mechanisms in different markets, and external forces. ZLB brings about higher real bond return as monetary policy fails to adapt to deflation. How does ZLB alter shock propagation depends on the resource flows induced among capital (private investment), bond and consumption. I conclude that ZLB is of limited concern and can be beneficial to the economy when its impact on the resource flows between capital and bond market is restricted, and aggregate employment is exogenous.

Introduction

ZLB has been an issue for monetary authorities in Japan, the US and the Eurozone for many years, monetary policy in Australia also had its first experience with the ZLB in 2020 following the onset of the COVID-19 pandemic. Though we have now departed from the low interest rate environment, the practical importance of analyzing economic dynamics when interest rate adjustment is restricted at ZLB has been emphasized once again. In this research, I aim to better understand the joint consequences of the ZLB for the macroeconomy and individual households.

ZLB tends to bind at severe recessions when aggregate demand is insufficient and deflation pressure exhausts systematic response of conventional monetary policy. It is therefore important to understand the resource flows in the economy that interacts with performance of different sectors. A well-known phenomenon in times of market stress is 'flight-to-liquidity' as investors reduce the risk exposure of their portfolios, generating aggregate resource flows from capital market to less risky and more liquid bond market (Beber et al., 2009). As bond ‘crowding out’ working capital available to firms, firms will necessarily adjust their behaviours in different aspects of their corporate operation, which is closely connected with employment status of households (labour market conditions) and their consumption-saving decisions. Meanwhile, government agents are free to exert forces on the economy, in the same way as exogenous aggregate shocks, trying to alter and improve the economic conditions.

The above narrative can be conceptualized into a trinity framework of macroeconomic analysis: 1) endogenous resource flows among capital, bond, and consumption 2) mechanisms in different economic sectors resource flows 3) external forces that can alter resource flows and dynamics in different market sectors. In general, resource flows are connected to aggregate supply and demand in the economy. Mechanisms in different economic sectors (agent/market) provides a vector field for resource flows. External forces can alter the vector field and influence resource flows across different uses, which closely interacts with economic mechanisms and feeds back into resource flows – change of the high-dimensional economic position in the vector field. Based on the framework, the factor of interest can be analyzed by studying its direct effect on the specified economic mechanisms and corresponding indirect effect as the deviating trajectory from the equilibrium point.

The primary implication of binding ZLB on resources flows is its alteration of real interest rate when central banks cannot respond to further deflation in the economy by lowering the short-term policy rate to negative. Higher real returns influence the behaviours of both lenders (households) and borrowers (the government). On the one hand, it becomes more attractive for households to save relative to consumption. This is the direct effect of ZLB on intertemporal consumption substitution of households. On the other hand, the government faces more costly bond repayment obligation. To maintain the budget balance, the government needs to consider either increasing taxation on household incomes or issuing more bonds. If fiscal policy cyclically responds to ZLB events, it necessarily affects the income stream of households, inducing changes of their behaviours indirectly. Much of the attention lies at understanding the relative importance of different channels at play, for they can indicate the effectiveness of different policy interventions. There are a range of associated channels that should be considered in a comprehensive model to grasp the intricacy of the mechanisms in the economy and offer precise assessment of the theories and beliefs (Colciago ,2019). With reference to the trinity framework, this paper contributes to the analysis of ZLB issue via the combination of three elements in the model.

Firstly, I extend the NK analysis of ZLB by considering the crucial role played by capital as a complementary production factor to labour and as an alternative saving instrument for households other than bond. Existence of capital market complicates the resource flows in the economy. Private investment in capital contributes to the aggregate demand apart from consumption, which solely determines the aggregate demand in NK models with no capital. It sets a stage for the model to capture 'flight-to-liquidity' phenomenon – resource flows from capital to bond market – at times of uncertainty, which is crucial in accounting for insufficient aggregate demand in addition to a reduction of consumption.

Secondly, different from the canonical HANK models with unrestricted state space of capital and bond choices, I employ an empirically inspired asset allocation rule that describes household preference of portfolio compositions as their wealth level varies, according to which the aggregate resource flows among capital, bond and consumption can be determined by optimal choices of households. However, the asset allocation rule restricts endogenous portfolio adjustment behaviours in the short run when holding the household wealth levels constant. I emphasize that it is important to devise a mechanism on interactions between capital and bond market based on household investment behaviours, which can help to capture the correct magnitude of endogenous resource flows and the corresponding propagation of aggregate shocks.

Thirdly, the model incorporates a reduced-form real wage rigidity to allow endogenously propagating aggregate shocks via household income dynamics, a role that cannot be served by nominal price adjustment cost. It allows the model to intrinsically generates lagged build-up of deflation pressure that leads to ZLB events (Blanchard & Gertler, 1999). Introducing real wage rigidity in the determination of marginal cost of production is necessary to activate price setting behaviours of firms if there is no further micro-foundation of mechanisms in capital and labour market that alters wage rate and capital return from the levels that would otherwise be determined by the marginal product of production factors in the firm problem. I prove the above statement and stress the point that, micro-foundations and imperfections that influence the determination of wage and capital return is important for real marginal cost to deviate from the optimal level.

The framework can be used to describe and compare to a wide class to micro-founded macroeconomic models that analyze forces and resource flows. The HANK model employed here does not offer comprehensive inspection of every aspect of ZLB because of muted shock propagation through labour market conditions and endogenous portfolio adjustment behaviours. Notwithstanding, I move a step forward by analyzing ZLB in the HANK framework, adding consideration of capital market and household heterogeneity, extending the analysis of ZLB in NK framework.

The abstraction of the model from employment environment and portfolio adjustment behaviours are closed related to the real wage rigidity and portfolio rule in the model. I use a reduced-form real wage rigidity to generate inflation dynamics in the model, which is similar to the form adopted by Blanchard & Galí (2007). For the flexible component of wage, the authors use the marginal rate of substitution between consumption and labour supply from the household problem to study the implication of real wage rigidity on welfare. Since labour supply is exogenous in the model, marginal product of labour is used to pinpoint the flexible component of wage. The rigid component depends on the wage in the previous period, making wage backward dependent. Real wage rigidity offers a mechanism of endogenous shock amplification in the economy, relaxing the dependence of the model on exogenous shocks in generating economic responses. If employment dynamics is endogenous, then the insufficient wage adjustment to aggregate shocks will necessarily imply response of changing labour demand, giving rise to the shock propagation channel in the labour market. By holding exogenous employment dynamics, I provide a counterfactual analysis of ZLB issue and imply significant amplification of aggregate shocks via labour market.

On the other hand, though introducing capital allows the study of implications of ZLB on the resource flows involving capital market, it imposes significant computational challenge on solving the model. The dimensionality issue can be resolved by realizing a pre-determined empirical portfolio rule of household investment that maps total wealth of households into capital and bond holdings. The empirical asset mapping is generated from Australian household balance sheet across the wealth distribution (ABS, 2023a). It captures the distinct risk feature of capital and bond and heterogeneous household preference of portfolio composition across the wealth distribution. The model does not rely on the liquidity conditions of different assets to account for risk premium of illiquid capital because of the cost associated with capital adjustment, which is crucial in simulating wealthy Hand-to-Mouth(HtM) households as they are so attracted by the high return of capital that they do not hold any liquid asset in their portfolios. By restricting the state space with the exogenous portfolio rule, the model differs from HANK with illiquid capital in the sense that the notion of wealthy HtM households does not apply in the model. Australian household data across the wealth distribution by quintiles (ABS,2023a) suggests that wealthy households, on average, are rational enough to hold certain fraction of liquid asset (bond equivalent) in their portfolio to facilitate their short-term consumption demand, which deviates from the account in Kaplan et al. (2018) and Auclert (2019). More detailed micro-level data on household balance sheet would be required to find out the relative size of wealthy HtM households.

The adoption of an exogenous portfolio rule implies that the model forfeits the ability to capture any short-run endogenous portfolio adjustment behaviours (Kaplan et al., 2018), which can be an important channel for shock propagation in the economy. The portfolio rule only allows changes in household portfolio composition to the degree of corresponding changes in total asset level, which impairs the interactions between capital and bond market. Notwithstanding the compromise on portfolio choices, the evolution of Australian household asset allocation across the wealth distribution in 21st century (ABS,2023a) indicates a persistent structural change of household portfolios to shocks that severely alters their risk preference of investment (See Appendix A.). Following the GFC in 2008, the experience and perception of higher risk in the capital market leads to a consistent portfolio adjustment towards higher weight on the bond (liquid) market. On the other hand, the shock of Covid-19 pandemic has not been observed to significantly alter household portfolios despite potential second-hand impact on capital and bond markets from lockdowns, fiscal stimulus and expansionary monetary operations. The observation suggests that extreme and direct impact on capital and bond market is required for households to materially adjust their asset allocation. The model can provide decent account of the position and exposure of households to aggregate shocks and their responses when the shocks do not exert direct impact on the risk involved in capital market and therefore portfolio composition largely remains invariant.

In this paper, I study aggregate and household dynamics at ZLB events generated by a negative total factor productivity (TFP) shock and positive preference shock respectively. TFP shock creates inflation initially due to higher production, but it is followed by a hump-shaped deflation response as continuous downward pressure on wage builds up due to real wage rigidity. Preference shock generates ZLB events by lowering the path of real marginal cost, leading to immediate deflation because of firm price setting behaviors. Results in both of the shocks suggest that ZLB is of limited concern if its impact on aggregate employment and resource flows in asset markets are restrained.

Literature Review

In essence, the evolution of micro-founded macroeconomic models can be described as continuous refinement of model specification of economic mechanisms such that more accurate description and prediction of empirical resource flows and impact of exogenous shocks can be obtained from the model. Many elements in HANK framework contribute to the model's ability to account for mechanisms in the economy.

HA component of HANK provides significant micro-foundation of the household problem by considering heterogeneity of households. It allows the model to specify a spectrum of household economic profiles and distinguish their behaviours as their economic status varies. This is important in capturing precise aggregate resource flows based on different consumption and investment decisions across households. While representative agent models had to conclude that intertemporal consumption is insensitive to interest rate changes to explain the empirically observed persistent level of consumption (Hall, 1988), the enrichment of household heterogeneity offers a satisfactory answer by accounting for the behaviour of constrained households who do not operate on the Euler's equation in the spirit of the spender-saver framework (Campbell & Mankiw, 1989). Moreover, household heterogeneity also has important implication on ZLB issue. Aiyagari (1994) modelled a fully-fledged HA model with idiosyncratic unemployment risk and incomplete market to explain the importance of precautionary saving incentives and its contribution to aggregate saving behaviors. In agreement with Huggett (1993), excess supply of capital and borrowing constraints lead to a steady state real interest rate that is lower than the implied natural rate from the discount factor. Precautionary saving behaviour of households makes HA economy intrinsically more prone to low-interest environment and occurrence of ZLB events. Subsequent literatures consider more dimensions of heterogeneity of households in terms of their discount factor of future (Krusell & Smith, 1998) and their varying portfolio composition (Kaplan et al. 2018; Hubmer et al. 2021), which allow more realistic depiction of household profiles and more precise construction of mechanisms in the household sector.

On the NK side, various features in NK framework are crucial in the analysis of monetary policy and business cycles. Among imperfections in price and wage level, the inertia of nominal wage plays a dominant role in generating the hump-shaped economic response to a monetary shock (Christiano et al., 2005). Real wage rigidity can also play the same role as nominal wage rigidity in accounting for the empirically observed lagged response of inflation (Galí&Gertler, 1999). The imperfections are incorporated into the firm problems in the model, producing inflation dynamics via price setting behaviour of the firms. By introducing the empirical imperfections, NK models add another layer of realism to mechanisms in the economy such as price setting behaviours of firms, employment conditions, and household employment, income and wealth dynamics.

However, an embedded difficulty facing NK model in studying ZLB phenomena is its linearizing technique, which imposes challenges on describing the nonlinear dynamics at ZLB. (Fernández-Villaverde et al., 2015). The main-stream NK literature offers solutions to the issue (Eggertsson & Woodford, 2003; Kulish et al., 2017). Other than the technical issue involved in NK model, the economy described by NK models is debatable because of the absence of capital market. In NK models with only bond market, higher real bond return caused by binding ZLB can bring out severe disruption to the economy because of the following properties: 1) the dominant role of bond in determining the portfolio return households perceive; 2) high sensitivity of consumption to changes in the path of real interest rate 3) equivalence of consumption to aggregate demand (Eggerston & Woodford, 2003; Galí, 2015). The combination of the first two properties gives rise to decline in consumption following an increase of real interest rate at ZLB as households find it more attractive to consume less and increase saving. The third property leads to the amplification of ZLB events – a reduction in consumption is translated to an equivalent decline in aggregate demand, generating further deflation pressure and higher real interest rate.

These properties are either relaxed or does not hold when capital market is considered. To start with, the direct effect of a change in bond return on the interest rate relevant to households depends on the share of bonds in their portfolios. Changes in bond return can only have a diluted impact on households if they have limited position in bond market. It would then require additional specification of the spillover effect of changes in bond return on capital return to gauge the net effect on the portfolio return that households perceive and the importance of intertemporal consumption sensitivity to interest rate changes. The second property is amended by introducing household heterogeneity via idiosyncratic employment uncertainty and incomplete insurance market. The possibility of hitting a borrowing constraint in the future effectively truncates the horizon of future valuation of households(Bilbiie, 2019; Kaplan et al., 2018), reducing their responsiveness to changes in the path of interest rate. For the constrained households, their consumption is naturally insensitive to interest rate changes, while responsive to changes in their income level. Hence, aggregate consumption level is more robust to high real rate environment caused by ZLB. Furthermore, capital also serves as an alternative saving instrument for households, complicating the determination of aggregate demand in a way that a sole reduction in consumption can be offset by increasing investment in capital. It removes the third property and makes the condition of decline in aggregate demand more stringent for an amplification of the deflationary pressure at ZLB.

Consequently, due to the extra degree of freedom introduced by capital, for ZLB to exert the same impact on the economy in the two-asset HANK as in NK models, there are two conditions need to be satisfied: 1) return on bond and capital are highly positively correlated; 2) private investment in capital replicates the movement of consumption. The first condition recovers the dominant effect of bond return on portfolio returns that households perceive as in NK models. If capital return is robust to or even offsets changes in bond return, then household consumption schedule will not be responsive at ZLB due to invariant portfolio returns. The second condition replicates the property in NK models without capital that consumption is equivalent to aggregate demand. When households reduce consumption to save more in face of exogenous shocks, a corresponding reduction of investment in capital is required for the economy to experience the same decline in aggregate demand, which amplifies deflation at ZLB. However, these two conditions are not guaranteed to hold in HANK models.

Many topics in NK literature wait to be revisited via HANK model. Taking a step forward is the study on the solutions to ZLB issue such as the potency of forward guidance in stabilizing economic conditions (Werning, 2015; McKay et al., 2016). Taking a step backwards is the analysis of various transmission channels of monetary policy shocks at the margin (Kaplan et al., 2018, Auclert, 2019). Standing in-between is the study of the nature of ZLB and its associated consequences on the economy. This paper is not the first effort on analyzing the implications of ZLB in HANK framework, Fernández-Villaverde et al. (2023) provide positive evidence on the amplifying effect of ZLB on negative demand shocks as monetary policy fails to accommodate to deflation. It is also concluded that precautionary saving behaviours induced by employment uncertainty makes the economy more vulnerable to occurrence of ZLB. The authors also emphasize how more costly price adjustment weakens inflation dynamics and naturally reduces the duration and frequency of ZLB events. However, the authors do not consider the role of capital in the HANK they employ, I contribute to this analysis with specific consideration of the role played by capital in both the production process and household saving behaviours.

The rest of the paper is structured as follows. Section I lays out the model. Section II introduces the calibration strategy and compare steady state properties of the model with data; Section III conducts quantitative analysis of ZLB events generated by a supply shock and a demand shock respectively and provide evidence on the importance of role played by real wage rigidity. Section IV concludes with an extended discussion from the results.

I. Model specification

There are four sectors in the model. Firstly, there is a final-good producer aggregating a continuum of monopolistically competitive intermediate-good producers. Intermediate firms minimize their cost of production given the demand from the final-good producer by choosing the amount of capital and labour to employ. They also affect the inflation dynamics in the economy by setting price today to maximize their profit. Secondly, households maximize their lifetime consumption by balancing consumption and saving today. Thirdly, the government distributes unemployment insurance, UI, to unemployed households and meets the debt repayment requirement by taxing total income of households and issuing new bond. Lastly, the central bank sets the nominal rate on bond in response to inflation in the economy according to the Taylor's rule. There are two distinct features in the model. The model incorporates real wage rigidity (Blanchard & Galí, 2007) in the NK component and specifies a portfolio rule for asset allocation in the household problem. These two extensions are crucial in the joint analysis of ZLB events and heterogeneous household behaviours.

I.A. Firms

I.A.1. Final-good producer

$$Y_t = \left(\int_0^1 Y_{t,i}^{1-\frac{1}{\epsilon}} di\right)^{\frac{\epsilon}{\epsilon-1}}$$

$$s.t. \int_0^1 P_{t,i} Y_{t,i} di = P_t Y_t$$

The final-good producer in the economy aggregates intermediate goods according to the demand schedule with constant elasticity of substitution (Dixit & Stiglitz, 1977). Given the aggregate budget constraint, solving the problem yields the demand of intermediate goods for an arbitrary firm i:

$$Y_{t,i}(p_{t,i}) = \left(\frac{p_{t,i}}{P_t}\right)^{-\epsilon} Y_t$$

I.A.2. Intermediate-good producers

There is a continuum of monopolistically competitive intermediate firms producing products as input for the final-good producer. Given the demand from the final-good producer, the steady state markup over real marginal cost can be conceptually computed from the following profit maximization problem:

$$\max_{p_{t,i}} \pi(p_{t,i}) = p_{t,i}Y_{t,i}(p_{t,i}) - C(Y_{t,i}(p_{t,i}))$$

$$\Rightarrow \frac{p_{t,i} - C'(Y_{t,i})}{p_{t,i}} = \frac{1}{\epsilon}$$

Given a unit price level, the steady state real marginal cost m^* is $\frac{\epsilon-1}{\epsilon}$.

The firms share the same production function with capital and labour as production input and access the same total factor technology (TFP), A. The production cost consists of capital return (r^k) and wage (w), which equal to real marginal cost (m) times the marginal product of capital and labour (MPK & MPN) respectively. Off-steady-state m is specified at the end of the firm problem. The firms distribute the monopolistic profit (π^M) to households equally after covering a quadratic price adjustment cost Θ^p (Rotemberg, 1982). The properties above can be described by the following equations:

$$Y_{t,i} = A_t {K_{t,i}}^{\alpha} {N_{t,i}}^{1-\alpha}$$

$$Y_{t,i} = r_t^k K_{t,i} + w_t N_{t,i} + \Theta_{t,i}^p + \pi_{t,i}^M$$

$$MPK = \alpha A K^{\alpha-1} N^{1-\alpha}$$

$$MPN = (1-\alpha) A K^{\alpha} N^{-\alpha}$$

The firms operate optimally by solving the following cost minimization problem and intertemporal profit maximization problem.

I.A.3. Cost minimization

$$\min_{K_{t,i},N_{t,i}} r_t^k K_{t,i} + w_t N_{t,i}$$

$$s.t. A_t K_{t,i}^{\alpha} N_{t,i}^{1-\alpha} = Y_{t,i}$$

Firms minimize the cost of producing an arbitrary amount of output, $Y_{t,i}$, by choosing the amount of capital and labour to employ given the capital return and market wage. Solving the problem produces the expression for m_t and the corresponding r_t^k & w_t:

$$m_t = \frac{1}{A_t}\left(\frac{r_t^k}{\alpha}\right)^{\alpha}\left(\frac{w_t}{1-\alpha}\right)^{1-\alpha}$$

$$r_t^k = m_t \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

$$w_t = m_t (1-\alpha) A_t K_t^{\alpha} N_t^{-\alpha}$$

I.A.4. Profit Maximization

$$\max_{p_{t,i}} V_{t,i} = Y_{t,i}\left\{\left(\frac{p_{t,i}}{P_t} - m_t\right) - \frac{\theta_p}{2}\log\left(\frac{p_{t,i}}{p_{t-1,i}}\right)^2\right\} + \beta V_{t+1,i}(p_{t,i})$$

$$s.t. Y_{t,i} = (\frac{p_{t,i}}{P_t})^{-\epsilon} Y_t$$

Profit maximization requires the choice of an optimal price level today given the demand from the final-good producer. Not only does the price matter for the revenue today, but it also plays a role in anchoring future prices due to the existence of the price adjustment cost (Rotemberg, 1982). Because the firm problem is symmetric, it can be solved to yield the following New Keynesian Phillips Curve (NKPC) with a quadratic term of small quantity involving inflation being ignored:

$$log\Pi_t = \beta \frac{Y_{t+1}}{Y_t} \log \Pi_{t+1} + \frac{\epsilon}{\theta_p}(m_t - m^*)$$

where $m^* = \frac{\epsilon-1}{\epsilon}$ is the steady-state real marginal cost derived from above.

I.A.5. Real Wage Rigidity

The final building block of the firm problem is the imperfection of real wage being rigid in the labour market. The formulation of real wage rigidity resembles the one in Blanchard & Galí (2007). Different from the authors, since household labour supply is exogenous in the model, the flexible part of wage is anchored by m^* times MPN_t. m^* is used instead of m_t because it makes the inflation dynamics computationally more tractable. It ignores further imperfection of real wage rigidity stemming from endogenous markup. The rigid component of wage is inherited from the previous period, given the wage rigidity parameter θ_w. As such, the equation of real wage is of the following form:

$$\widetilde{w}_t = \theta_w \widetilde{w}_{t-1} + (1-\theta_w) m^* (1-\alpha) A_t K_t^{\alpha} N_t^{-\alpha}$$

Real wage rigidity plays an important role in generating inflation dynamics when dynamics in the labour and capital market are restricted to only the firm problem described above. The necessity of this imperfection comes from the fact that – if both r^k and w are perfectly flexible as m^*MPK_t & m^*MPN_t, then real marginal cost is invariant over time and there are no inflation dynamics in response to shocks.

Proposition 1:

If $r^k = \lambda MPK$ and $w = \lambda MPN$, where λ is an arbitrary constant, then m is invariant over time and equals to λ.

Proof: Let's firstly express m in coordinate (A, r^k, w):

$$m(t, A, r^k, w) = \frac{1}{A}\left(\frac{r^k}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}$$

Taking the total time derivative:

$$\dot{m} = -\frac{1}{A^2}\left(\frac{r^k}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}\dot{A} + \frac{1}{A}\left(\frac{1-\alpha}{\alpha}\frac{r^k}{w}\right)^{\alpha-1}\dot{r^k} + \frac{1}{A}\left(\frac{1-\alpha}{\alpha}\frac{r^k}{w}\right)^{\alpha}\dot{w}$$

r^k and w can be equivalently expressed in cartesian coordinate (A, K, N) based on $r^k = \lambda MPK(A, K, N)$ and $w = \lambda MPN(A, K, N)$. That is,

$$\dot{r^k} = \lambda\alpha K^{\alpha-1}N^{1-\alpha}\dot{A} + \lambda\alpha(\alpha-1)AK^{\alpha-2}N^{1-\alpha}\dot{K} + \lambda\alpha(1-\alpha)AK^{\alpha-1}N^{-\alpha}\dot{N}$$

$$\dot{w} = \lambda(1-\alpha)K^{\alpha}N^{-\alpha}\dot{A} + \lambda\alpha(1-\alpha)AK^{\alpha-1}N^{-\alpha}\dot{K} + \lambda(-\alpha)(1-\alpha)AK^{\alpha}N^{-\alpha-1}\dot{N}$$

Transfer m into cartesian coordinate (A, K, N) and simplify the expression:

$$\dot{m} = -\frac{1}{A}\left(\lambda\dot{A} - \left(\frac{N}{K}\right)^{\alpha}\lambda K^{\alpha}N^{-\alpha}\dot{A}\right) = 0$$

Hence, m is invariant over time given an arbitrary constant λ. ■

Proposition 1 is applied to the firm problem when λ is equal to $m^* = \frac{\epsilon-1}{\epsilon}$. Consequently, there is no inflation in the economy as can be depicted by NKPC when $m_t = m^*$. To be more specific, when aggregate forces lead to changes in $A, K, and\ N$, steady-state real marginal cost is conserved and price adjustment cost is avoided because perfectly flexible production cost of capital and labour completely negate changes in TFP, and diminishing return of capital/labour is offset by its complementary effect on the other production factor. As such, additional imperfection in either capital or labour market is required along with the nominal price adjustment cost to generate inflation dynamics.

The current wage level can then be expressed as the accumulation of shock impact on the previous path of MPN:

$$\widetilde{w}_t = \sum_{i=0}^{\infty}\theta_w^i(1-\theta_w)m^* MPN_{t-i}(A_{t-i}, K_{t-i}, N_{t-i})$$

The further the deviation it is in the past, the less powerful it is in influencing the wage level today. As such, by incorporating the real wage rigidity, real marginal cost equation and NKPC can now be expressed as

$$m_t = \theta_w \frac{\widetilde{w}_{t-1}}{MPN_t} + (1-\theta_w)m^*$$

$$log\Pi_t = \beta \frac{Y_{t+1}}{Y_t} \log \Pi_{t+1} + \frac{\theta_w \epsilon}{\theta_p} (\frac{\widetilde{w}_{t-1}}{MPN_t} - m^*)$$

It is clear that the rigidity in wage is transmitted to both the real marginal cost and inflation. It is further passed down to r^k and r^b, contributing to a lagged peak of shock impact overall (hump-shaped aggregate dynamics). It agrees with the views in Galí & Gertler (1999) that wage rigidity can be translated to sluggish real marginal cost and account for slow response of inflation to output gap and real wage, as the real unit labour cost shares contemporaneous dynamics with inflation. In steady state, m^* is recovered when lagged real wage does not contribute to any deviation of real marginal cost and inflation is zero in the economy.

I.B.1. Households

Infinitely-lived households maximize their life-time consumption by optimizing the choice of consumption c and asset holdings (a) – consisting of capital (k) and bond (b) – according to the following recursive form value function.

$$V_t(k_t, b_t, z_t) = \max_{a_{t+1}, c_t} u(c_t) + \beta E_t[V_{t+1}(k_{t+1}(a_{t+1}), b_{t+1}(a_{t+1}), z_{t+1})]$$

subject to:

$$c_t + k_{t+1} + b_{t+1} = [w_t z_t + UI\mathbb{1}\{z_t = 0\} + r_t^k k_t + r_t^b b_t](1-\tau_t) + k_t + b_t + \pi_t^M$$

$$a_t \geq -\frac{UI}{1 + E_t[r_{t+1}^b]}$$

Households receive after-tax $(1-\tau)$ income from wage, unemployment insurance *UI* if unemployed *z=0*, and returns on capital and bond (r^b). They also receive monopolistic profit (π^M) from firms uniformly. The labour income that households receive also depends on their current employment status (z), which is governed by a three-state Markov chain (See Calibration table), including unemployment, average employment, and superstar stage (Rosen, 1981). It is the source of heterogeneity in the model. For instance, households that keep the prime employment status can accumulate wealth steadily subject to a drop to average employment status. As such, households form expected value of future given this employment uncertainty. Moreover, households can borrow up to the amount of unemployment insurance discounted by the expected bond rate to increase consumption. It is a natural level of borrowing constraint because households who need to borrow don't have enough income and wealth to support their current consumption (there is no liquidity constraint on assets), and UI is the worst income level households can receive in the next period to repay the debt.

Since households can choose to invest in both capital and bond market, the extra degree of freedom imposes significant computational challenge on solving the model. A trick to reduce the degree of freedom is to assume a predetermined asset allocation rule for different wealth levels. It is a mapping from the total asset level of households to the optimal level of capital and bond to hold. Household holding of capital and bond in their portfolios is determined by the following rules:

$$\mathcal{K}(a) = \begin{cases} \dfrac{1}{ceil + e^{-\gamma(a-mid)}} + floor & if\ a \geq 0 \\ 0 & if\ a < 0 \end{cases}$$

$$\mathcal{B}(a) = 1 - \mathcal{K}(a)$$

The first piece of the portfolio rule is modified from the logistic function. It is monotonically increasing in terms of households' wealth level. $ceil$ restricts the ceiling of weight on capital; mid is the middle point of the mapping where the total asset corresponds to the midpoint of the achievable range of w^k; $floor$ controls the lower bound of w^k; γ determines the smoothness of the transition of optimal weight on capital as wealth level changes. Such approach captures the exogenous risk factors involved in capital and the varying risk preference of households with different wealth levels.

I.C.1. Government and Fiscal Policy

The government faces the following budget constraint:

$$(1-\tau_t)UI\Lambda(\mathrm{z}=0) + \frac{1+i_t}{1+\pi_t}B_t = \tau_t\left(\widetilde{w}_t N_t + r_t^k K_t + r_t^b B_t\right) + B_{t+1}$$

The government distributes unemployment insurance to unemployed households, where $\Lambda(z=0)$ is the cumulative distribution function of employment status up to $z=0$. The government also fulfills its debt repayment obligations using revenues from taxation of household income and bond issuance. Since bond supply adapts to bond demand from households, when ZLB is binding due to strong deflationary forces in the economy, it becomes more costly for the government to repay the existing debt. Since government bond plays a role as an instrument of saving and thereby facilitating consumption, aggregate level of bond is endogenously responding to the nominal rate set by the central bank in the interest-setting regime. Consequently, more taxation of household income is required to cover the additional expenses incurred by ZLB for the budget clearing condition to be satisfied. Though the government follows this structural and stringent principle, implications from the model still apply to more realistic fiscal policy operation when more instruments and flexibility are provided.

I.D.1. Central Bank and Monetary Policy

$$i_{t+1} = \begin{cases} r_n + \phi_\pi \pi_t & if\ \pi_t \geq -\dfrac{r_n}{\phi_\pi} \\ 0 & if\ \pi_t < -\dfrac{r_n}{\phi_\pi} \end{cases}$$

The central bank responds to inflation in the economy by setting the nominal rate based on a basic Taylor's rule (Kaplan, 2018) and faces the ZLB constraint. With an inflation target of 0, The implied binding condition of ZLB is when inflation becomes lower than $-\frac{r_n}{\phi_\pi}$.

I.E.1. General Equilibrium

Solving the model requires the following general equilibrium conditions: 1) capital supply from households is equal to capital demand of firms according to the capital return. 2) Aggregate bond supply is equal to bond demand of households in their portfolio. 3) Budget constraint of the government needs to be satisfied by adjusting the aggregate tax revenue. 4) Aggregate output produced by the firms must equal to the sum of aggregate consumption, private investment in capital, price adjustment cost, and the monopolistic profit.

When solving for a time path of general equilibrium economic response to a shock, I assume perfect foresight of households by iterating their value function backwards along the time path of the exogenous shock and guessed response of aggregate variables. According to the decision rules, I solve the household problem forward to update the time-path of prevailing aggregate variables. The iteration process is complete when state variables reach the fixed point (See Appendix B.).

II. Model Calibration and Properties

Table 1: Parameter Calibration

	Description	Value	Source/Target
Households			
σ	intertemporal risk aversion	2	McKay et al. (2016)
β	Discount factor	0.987	5.5% Annual depreciation rate
z	Idiosyncratic productivity	{0,1,3}	Primary Income ratio[1] (ABS,2023a)
Portfolio Rule			
ceil		6.5	
slope	Portfolio Rule	0	Household total asset composition
mid	Parameter Specification	30	by wealth quintiles (ABS,2023a)
floor		0.65	
Markov Chain			
s	Separation rate	5%	Internally calibrated
f	Finding rate	80%	
u	Unemployment rate	5.38%	eq'm unemployment rate (ABS,2023b)
Firms			
δ	Depreciation rate	0.18	8% annual depreciation rate
A	Total factor productivity	1	Normalized
α	Capital share	0.33	Kaplan et al. (2018)
ϵ	elasticity of demand	20	Kaplan et al. (2018)
θ_p	price adjustment cost	100	Standard Value
θ_w	Real wage rigidity parameter	0.7	Internally calibrated
Government			
τ	uniform tax rate		Steady state gov. budget clearing
UI	Unemployment Insurance	0.50	Secondary income ratio[2] (ABS,2023a)
Central Bank			
ϕ_π	responsiveness to inflation	1.5	Standard Value
r_n	natural rate	0.5%	Kaplan et al. (2018)
$\delta K/Y$	Consumption of fixed capital to output	0.182	0.188 (ABS,2023c) capital

The model is of quarterly frequency. Table 1 shows the calibrated parameters. Household-related parameters including heterogeneity of labour income, z, the optimal portfolio rule, unemployment insurance and unemployment rate are calibrated to Australian household data from ABS.

[1] Primary income of 5th quintile households / Average primary income of 2nd to 4th quintile households
[2] Average primary income of 2nd to 4th quintile households / Secondary income of 1st quintile households

Table 2: Portfolio Composition of Households across Wealth Distribution, by Quintiles

Wealth Quintile	1st	2nd	3rd	4th	5th
Weight on Capital	69.7% (70.4%)	72.2% (73.5%)	77.9% (78.4%)	80.2% (79.7%)	28.4% (84.3%)
Weight on Bond	30.3% (29.6%)	27.8% (26.5%)	22.1% (21.6%)	19.8% (20.3%)	19.6% (15.7%)

Table 2 compares the portfolio composition of households across the wealth distribution generated by the model and the Australian data for 2017/18 (ABS,2023). I exclude land from household balance sheet and define assets other than deposits and securities as capital. Because liabilities are largely loans and placements in the long run, I subtract liability from total asset to compute net wealth. The first quintile households have around 30% of their wealth invested in the bond market. As the wealth level increases, the weight on capital increases to 80% in the fifth quintile. The rising share of capital for households with higher wealth level indicates their preference for higher return and better ability to manage the short-term volatility involved in the capital market. Degree of asset liquidity is not distinguished as both capital and bond are considered liquid in the model, and they only differ in terms of their returns and the exogenous risk component. The assumption here is that households do not hold extreme portfolios such as 100% weight on capital (potentially illiquid) when it has higher return due to the exogenous preference of portfolios with different return, volatility, and liquidity depending on their wealth levels. The approach does not allow for independent portfolio adjustment behaviours to total wealth level and therefore, weakens the interaction between capital and bond market, it is still valid in pinpointing the cross-sectional exposure of households to capital and bond market.

Table 3: Distribution of Household Consumption, Income and Wealth, by Share of Aggregate

Wealth Quintile	1st	2nd	3rd	4th	5th
Consumption Share	14.3% (11.9%)	16.2% (14.2%)	19% (18.2%)	22.1% (22.3%)	28.4% (33.4%)
Income Share	10.2% (7.9%)	12.1% (12.3%)	16.9% (16.5%)	23.3% (21.7%)	37.4% 41.6%
Bond Share	1.2% (10.5%)	4.9% (13.1%)	13.4% (15.6%)	25.3% (18.9%)	55.2% (41.9%)
Capital Share	0.7% (6.0%)	3.3% (8.7%)	12.1% (13.6%)	26.2% (17.9%)	57.8% (54.0%)

Note 1: (·) uses Australian household data in 2018/19 (ABS,2023a)

Before diving into the thought experiments, it is important to know the different roles heterogenous households play for the aggregate economy (Rosen, 1981). Table 3 compares the distribution of aggregate variables across the wealth distribution with Australian household data (ABS,2023a). Consumption and income distributions are less right-skewed than aggregate capital and bond. Lowest quintile households account for sizable share of aggregate consumption (14.3%/11.9%), while only holding infinitesimal amount of bond and capital (1.2%/10.5% & 0.7%/6.0%)

III. Shock Experiments and Results

I analyze ZLB events generated by supply and demand shocks respectively. The first candidate is a 5% unanticipated TFP shock (supply side) following an AR(1) process with persistency of 80% is considered. The second shock is a 10% unanticipated preference shock on the discount factor of households (demand side) with 80% persistency. A set of counterfactual scenarios are produced to study the roles played by ZLB and real wage rigidity. Corresponding heterogeneous household dynamics is inspected alongside different scenarios to understand different household response across the wealth distribution.

III.A. Supply shock

III.A.1 Baseline response: ZLB & real wage rigidity

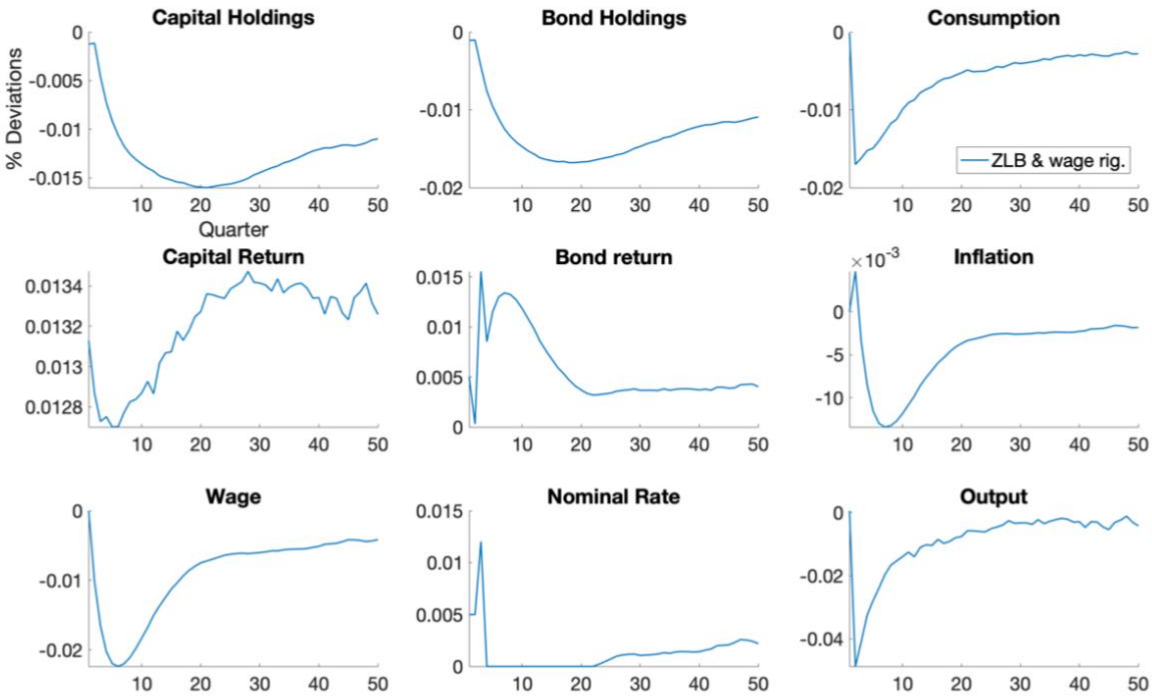

Figure 1: IRF to TFP Shock, Baseline Case

Figure 1 plots the IRF of the economy to the TFP shock in the general case with both ZLB and real wage rigidity. The initial response to a lower TFP is from firms. As they become less productive, they distribute lower wage and capital return to households. With lower total income level in general, households smooth consumption by running down their holdings of capital and bond simultaneously according to the portfolio rule, which feeds back into the production process, making labour even less productive and fueling further decline in real wage. Due to the presence of real wage rigidity, wage cannot efficiently be lowered to match the sharp decline in MPN immediately after the shock, making production more costly for firms. The increase in real marginal cost is reflected on the initial spike of inflation as firms face an increasing real marginal cost. Following the spike of inflation, as the downward forces accumulate in wage and the MPN recovers, the wage markup exceeds the steady state level $\frac{\epsilon}{\epsilon-1}$, which is reflected on a decline of real marginal cost and the corresponding persistent deflation.

Since the central bank is constrained by the ZLB in responding to the build-up of deflation with a negative policy rate, nominal rate stays at 0 for around 15 quarters and real bond return starts to increase following the deflation in the economy. The trend in capital stocks and the exogenous TFP shock jointly result in a sharp decline of output and a slow recovery because of the feedback loop between real wage rigidity and household behaviours – wage is persistently lower and households with lower total income invest less in capital market to maintain their optimal consumption schedule.

Figure 2: Household Heterogeneity in TFP Shock, General Case

Figure 2 compares the corresponding heterogeneous household dynamics across the wealth distribution by the lowest (1st), middle (3rd), and the highest quintile (5th) households. The lowest quintile households reduce their consumption by the most following the aggregate shock. For households with larger wealth holdings, they reduce their consumption by less amount as they have sufficient wealth to smooth consumption. In combination with Table 3, first quintile households contribute to 0.64 basis point of reduction in aggregate consumption while fifth quintile households contribute to only 0.17 basis point. The results suggest that poorer households are disproportionally more exposed to the negative supply shock. Likewise, they also experience a larger and more persistent decline of their total income and wealth as they are more constrained in smoothing their consumption.

It has been observed in Figure 1 that deflation leads to higher real bond return at ZLB. The additional profitability of bond is only beneficial to wealthy households with large enough position in the bond market as their income and wealth exceed the steady-state level when ZLB is binding. For poor households with limited wealth, they only perceive the relative increase in profitability of saving but need to run down their bond holdings significantly to smooth consumption, which prevents them to gain from the bond market.

III.A.2. The role of ZLB in the TFP Shock

To study the role of ZLB in detail, I decompose the twin effects of higher real bond return on lenders and borrowers at ZLB by comparing the baseline economy in V.A.1 with two counterfactual scenarios: 1) monetary policy is not constrained by ZLB and negative nominal rate is possible; 2) ZLB is in place but the endogenous response of fiscal policy to more costly outstanding debt is muted by assuming availability of external funds, removing the cyclical effect induced by ZLB.

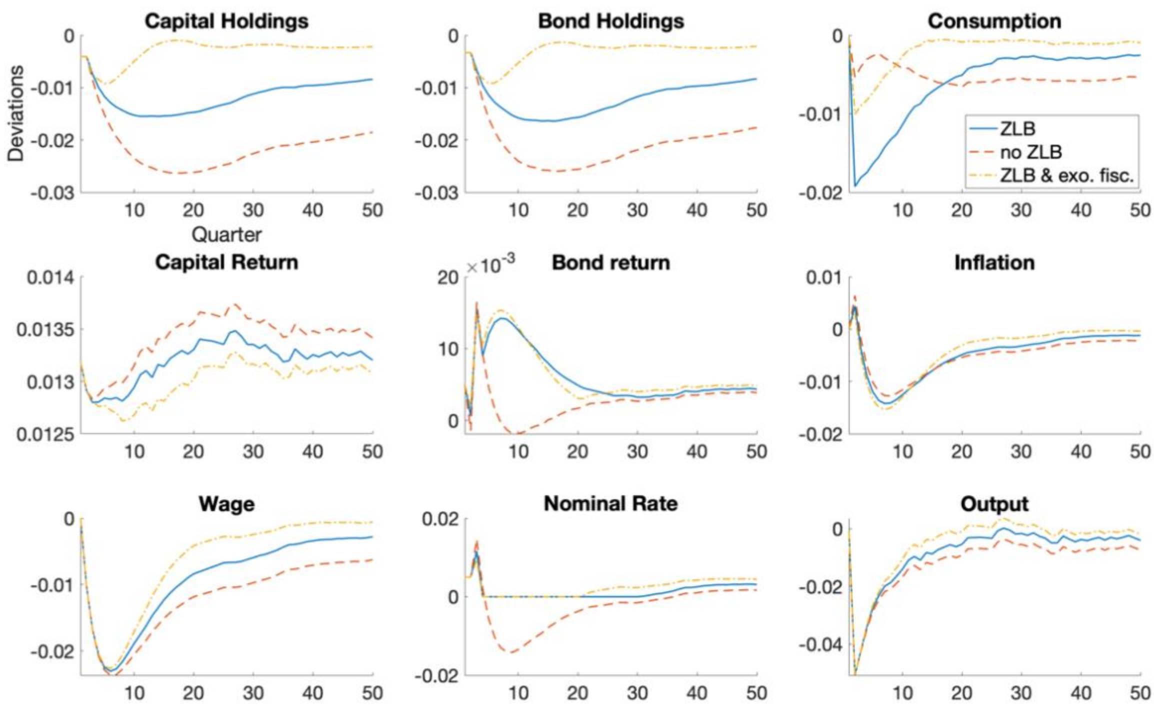

Figure 3: IRF to TFP Shock, Decomposition of ZLB

The primary difference between the economies with and without ZLB lies at central bank's ability to adapt to deep deflation. When ZLB is not imposed, nominal rate drops to negative, following the trend of deflation. Its immediate implication is the corresponding decline of real bond return in contrast to the increasing return when ZLB is binding. As such, households perceive a lower bond return in the future and have less incentive to save, which is reflected by a smaller decline of consumption and a larger decline in both aggregate capital and bond holdings. Although rigid wage inherits past deviation of TFP and capital from the steady state level, the additional drop in capital stock has a larger direct effect on the contemporaneous MPN, making production marginally more costly than that when ZLB is binding, as captured by a less severe trough of deflation. However, deflation is prolonged when ZLB is absent due to persistently lower wage when capital stock is also on a lower path.

The role of endogenous fiscal response to ZLB can be assessed by the third economy where the government can interact with an external sector and therefore sustain an exogenous fiscal policy. In this scenario, households do not expect additional taxation in the future when ZLB is binding. When households perceive higher disposable income in the future, they have less incentive to save now to guarantee future consumption, resulting in an intermediate drop of consumption as bond return will still increase at ZLB. Because the tightening fiscal response is avoided, capital and bond holdings efficiently recover to the steady state level instead of being dragged into persistently lower state if the government sustains its budget balance using domestic resources as in the first economy.

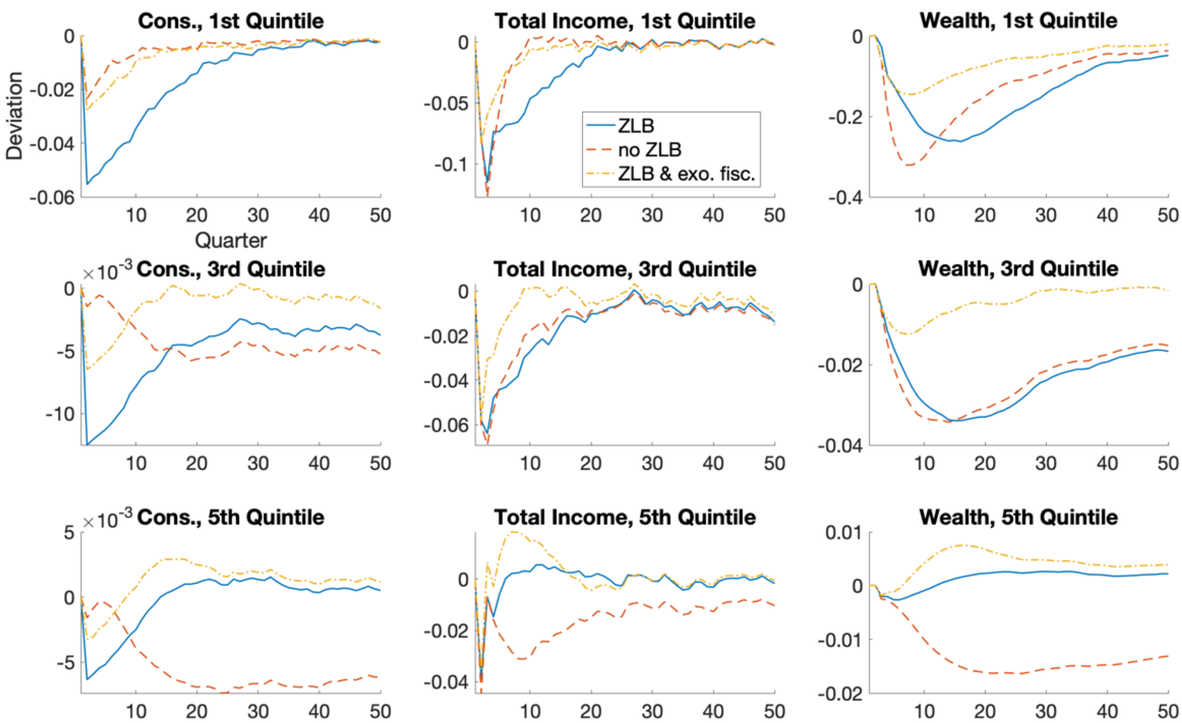

Figure 4: Household Heterogeneity in TFP Shock, Decomposition of ZLB

In Figure 4, I analyze the different sensitivities of households across the wealth distribution to the twin effects of higher real bond return generated by ZLB. From the economy without ZLB constraint (no ZLB) to binding ZLB with solely a higher path of bond return (ZLB & exo. fisc.) and then to the economy with endogenous fiscal response (ZLB) sequentially, more incentive to save is provided for households with different wealth levels. Lowest quintile households are not sensitive to a higher path of bond return since they are constrained in consumption and do not assign high enough value for the future, while more significant relative reduction in consumption is observed for middle and highest quintile households. Compared with the alteration of the path of bond return, endogenous fiscal response induces a more uniform increase in saving across the wealth distribution.

For the first quintile households, in the baseline economy with ZLB, the trough of their wealth level is delayed to the point where ZLB is binding, and the government is taxing more to fulfill its debt obligations, distinct from a sharp decline of wealth to smooth consumption in the case without ZLB. That is, instead of trying to profit from higher bond return, constrained households are incentivized to save more to buffer the expected tax rise at ZLB when deflation builds up. Additionally, since households borrow completely from the bond market, an expectation of higher cost of borrowing also leads to more precautionary savings to hedge the risk of being exposed to the costly debt in the future. Further evidence can be seen from the third economy where the fiscal policy is exogenous to ZLB. When ZLB does not induce fiscal tightening, the additional saving incentive provided by higher bond return can be observed in the third economy and the trough of wealth is reached simultaneously as in the no-ZLB economy for consumption smoothing without any delay to cover future rise in tax.

The results suggest that ZLB significantly benefits wealthy households because of the higher path of real bond return. They are less sensitive to the fiscal response as the gain from interest of bond is larger. When ZLB is not binding and therefore the central bank can set a negative rate to prevent an increase of real bond return, less profitability of bond market leads to a sizable decline of the income and wealth level of the fifth quintile households. Consequently, their consumption is also persistently lower because they face smaller marginal utility of consumption with a high level of consumption.

In relation to Auclert (2019), constrained households with higher MPC are losers of binding ZLB, dampening the efficacy of further operations on interest rate expectation management in stimulating aggregate consumption and calling for actions of fiscal policy. If tax progressivity is implemented and fiscal response works towards negation of adverse distributional effect on households, welfare of wealthy households (winners) can be transferred to constrained households (losers). It presents a trade-off for policy makers as, by worsening the profitability of holding wealth, the dominant position of wealthy households in contributing to aggregate capital stock predicts marginally poorer dynamics of the aggregate economy as their incentive to invest is impaired. Instead of conducting expectation management on future interest rate via forward guidance, a credible expectation management of future fiscal policy is found to be effective in stimulating aggregate demand. That is assuming feasible fiscal gains from external interaction in an open economy setting.

Moreover, due to the tangled asset allocation in capital and bond, an increase of incentive to consume that stems from expansionary interest management operations will lead to less investment in both capital and bond, negating the initial stimulatory effect of consumption on aggregate demand. Therefore, it provides evidence that NK model with no capital can exaggerate the efficacy of policies that operate through alteration of the path of interest rate. Consider abandoning the pre-determined portfolio rule, for policies like forward guidance to be effective in stimulating the economy, it would require the declining bond return to incentivize more private investment in firms – closing the negative output gap, which is at odd with the desired conditions in a preference shock for NK models as will be described in the second experiment. To complete the conjecture, if it is the opposite case that low interest return in bond market also leads to withdrawal from capital market as described in the model, then interest rate manipulation also contributes to a larger negative output gap.

III.B. Positive Preference Shock

In the second experiment, I conduct the same analysis on a ZLB event generated by a positive preference shock on time discount factor of households.

III.B.1. Baseline response: ZLB & real wage rigidity

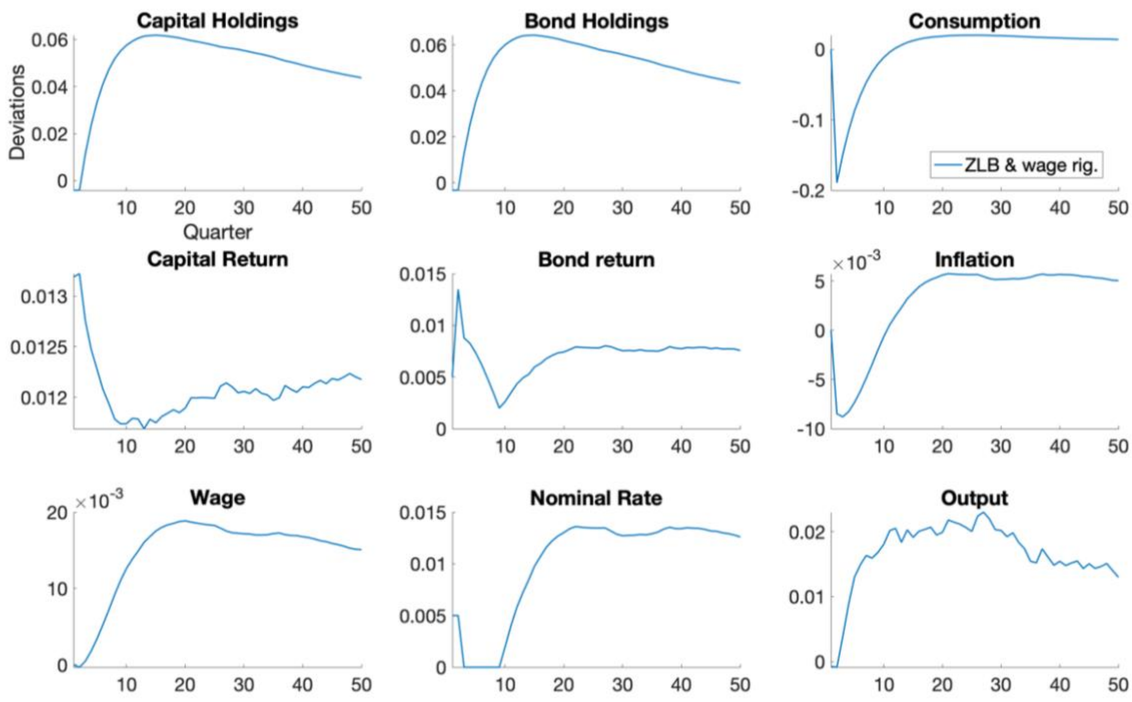

Figure 5: IRF to Preference Shock, Baseline Case

Figure 5 plots the baseline aggregate economic response to the positive preference shock. Since households put more value on the future stream of consumption, they reduce consumption today and save more in both capital and bond markets according to the asset allocation rule. Higher capital stock leads to the decline in capital return and a lagged increase of wage due to real wage rigidity. In expectation of lagged wage adjustment, firms immediate lower the price to profit from lower real marginal cost in the future, dragging the economy to ZLB environment. As firms expect the upward pressure to build up in the wage level, they start raising the price level continuously, ending up with persistently higher inflation in the economy as wage is difficult to adjust downwards.

Due to immediate realization of the trough of inflation, real bond return also reaches the peak when ZLB is binding. Further weakening of deflation pressure leads to a gradual decline of real bond return towards the steady state level. In response to the subsequent rising inflation that drives the economy away from ZLB constraint, the central bank is able to set higher nominal rate to stabilize inflation in the economy, creating higher real bond return. As the preference shock vanishes, households gradually recover their consumption level. Because of persistently higher real wage, households can consume more and save more, which slows down the convergence of the economy back to steady state level. As a result of higher capital stock throughout the shock experiment, output and inflation also follow the trend of capital, exhibiting a persistent positive output gap and inflation in the economy.

Figure 6: Household Heterogeneity in Preference Shock, Baseline Case

Figure 6 inspects the heterogeneity among household dynamics during the preference shock. Different from the TFP shock, wealthy households reduce consumption more in comparison with low-wealth households. This is because poorer households have lower level of consumption and therefore higher marginal propensity to consume (MPC). Wealth of lowest quintile households is almost doubled at the peak but quickly run down, while wealthy households sustain the positive gap of wealth from the steady state level. Since they hold the majority of wealth in the economy, the pattern of their wealth holdings plays the dominant role in determining the trend of aggregate capital and bond. Lastly, since a large fraction of the total income of wealthy households come from their investment in capital and bond markets, their income level plummets when capital return and bond return face a simultaneous decline – the joint result of universal increase of wealth across households and recovering inflation.

III.B.2. The role of ZLB in the Preference Shock

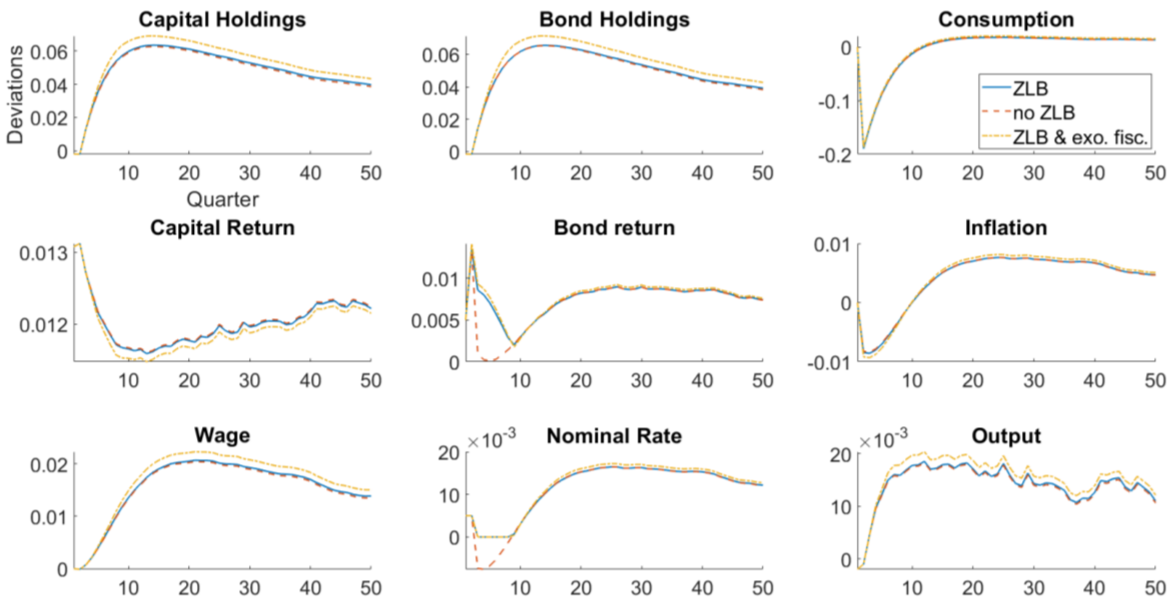

Figure 7: IRF to Preference Shock, Decomposition of ZLB

Figure 7 repeats the analysis of the decomposed effect of ZLB by considering the two other counterfactual scenarios. It can be concluded by reference to the economy without ZLB constraint that response of aggregate economy is robust to the impact of ZLB. Because the path of real bond return is lower when ZLB is not binding, incentive to save is marginally lower, resulting in negligibly lower path of aggregate capital and bond. In this case, monetary policy being able to adapt to inflation dynamics can help reduce the deviation of the economy from the steady state.

The implication of ZLB on the economy in a positive preference shock is in contrast with its effect in a negative TFP shock – higher real bond return induced by ZLB fuels the positive output gap in the former case but dampen the negative output gap in the latter. The exact influence can also be applied to the endogenous fiscal tightening in response to higher bond repayment cost at ZLB. In the preference shock, as total income level becomes higher, the government runs at a surplus when maintaining exogenous fiscal policy (no changes in the tax rate). If the government does not impose the cyclical lumpsum tax induced by ZLB on households, disposable income of households uniformly increases, which leads to more saving in capital and bond and contributes to a larger positive output gap.

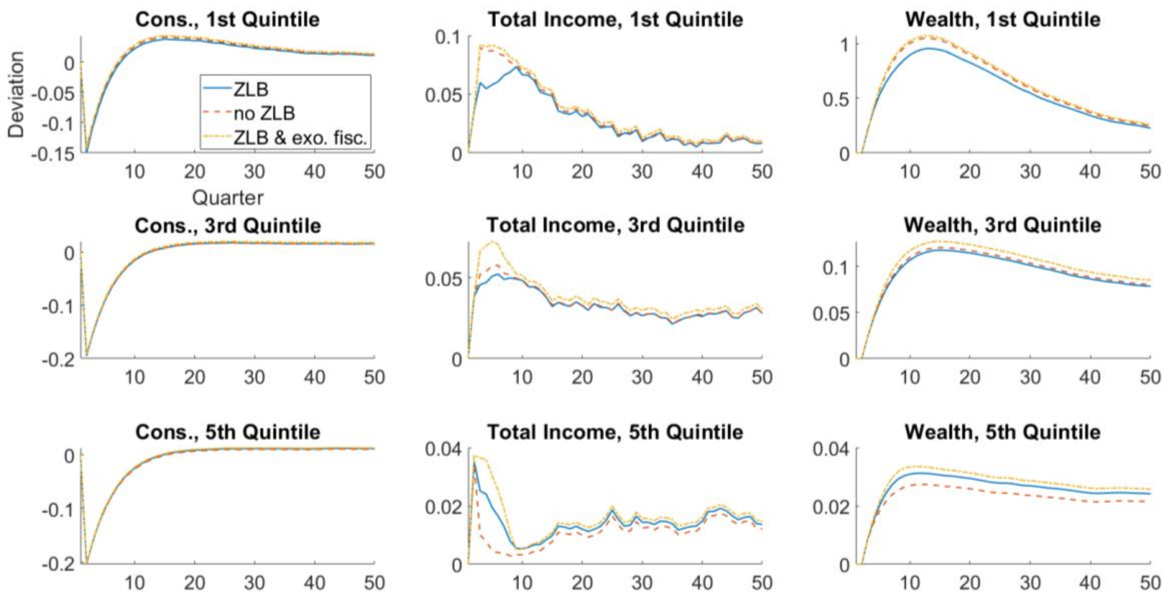

Figure 8: Household Heterogeneity in Preference Shock, Decomposition of ZLB

The corresponding household heterogeneity depicted in Figure 8 reenforces the fact that households across the wealth distribution have different sensitivities to fiscal response and the change in real bond return. For lowest quintile households, their total income and wealth are the highest when fiscal policy is neutral to ZLB. It is also evident that a sole increase in real bond return does not result in observable change of their income and wealth status by comparing the second and the third economy. As the wealth level of household increases, they gain more from higher bond return at ZLB, apart from the benefit of exogenous fiscal policy. It is worth noting that the consumption schedule of households across the wealth distribution are all robust to the alteration from ZLB, indicating the dominant effect of discount factor on household consumption.

Different from the results of standard NK models, household consumption does not directly affect inflation dynamics by determining the aggregate demand. The decline in aggregate consumption is majorly offset by the increase of investment in capital as the preference shock does not directly imply weak performance of capital market. This result is in stark contrast with the narrative of a positive preference shock in NK models. In HANK, the contribution of consumption on inflation dynamics following a preference shock operates through the production process and labour market by influencing the capital available to firm production and therefore the wage dynamics, which leads to a persistent deviation of the path of real marginal cost from steady state that dominates the pricing behaviour of firms (NKPC).

For a preference shock to reduce consumption and aggregate demand at the same time in HANK as in NK models with no capital, it would require a contemporaneous decline of investment in capital market, which cannot be directly implied by an exogenous positive discount factor shock – future consumption is more valuable than present consumption.

Here I complete the inspection of the efficacy of interest rate management policies in HANK and NK. In the preference shock, the model specifies strict co-movement of capital and bond because of the exogenous portfolio rule. It falls under the assumption that lowering bond return will lead to withdrawal from both capital and bond market (strong spillover effect to capital market) – an undesirable property in a negative TFP shock, where negative output gap can be exacerbated. However, in the ZLB event generated by the preference shock, strong positive connection between capital and bond market is desired as destruction of interest return expectation can discourage investment in both capital and bond market and mitigate the drastic drop of consumption. If expansionary interest management policies that lower the path of bond return leads to more investment in capital (the desired condition in a negative TFP shock), it will exacerbate the positive output gap and create stronger deflationary pressure as capital stock increases.

III.C.1. The role of Real Wage Rigidity in Inflation Dynamics

I have examined the ZLB issue in HANK model and emphasized the significance of modelling capital market. Throughout the analysis, real wage rigidity is fixed as an exogenous imperfection. Here I discuss the role it plays in generating inflation dynamics by adjusting the rigidity parameter. The baseline 5-quarter wage rigidity (0.8) is compared with two alternative specifications, 4-quarter wage (0.75) and 8-quarter wage (0.875).

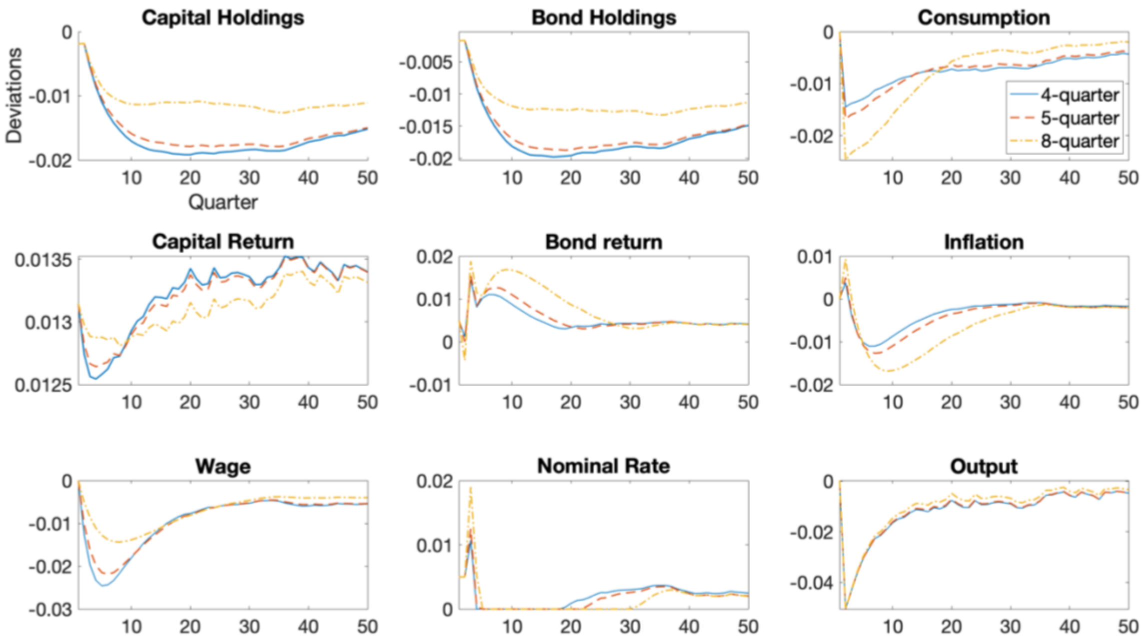

Figure 9.A: IRF to TFP Shock, Sensitivity to Real Wage Rigidity

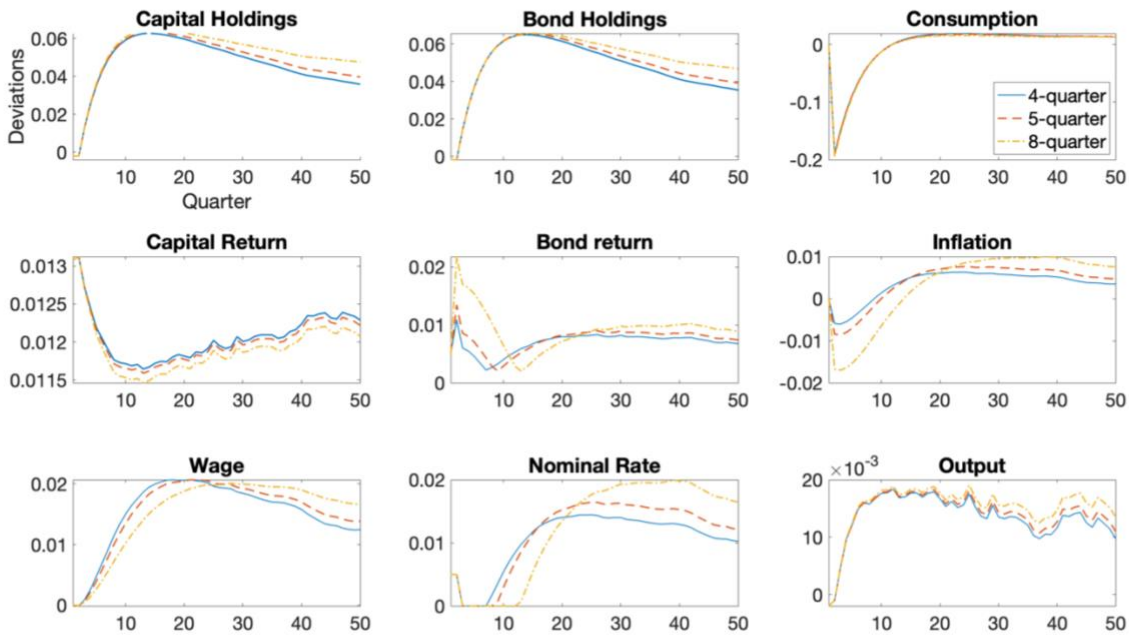

Figure 9.B: IRF to Preference Shock, Sensitivity to Real Wage Rigidity

Figure 9 shows that, in both the negative TFP shock and the positive preference shock, real wage rigidity is positively associated with the intensity of inflation dynamics. When real wage becomes less responsive to change in MPN, the deviation is accumulated in the real marginal cost, which is reflected on the general price level that firms charge. Therefore, the backward-dependent nature of the real wage rigidity endogenously amplifies the exogenous shocks and prolong the binding period of ZLB by creating lagged response of the economy in general. This is in contrast with the implication of nominal price rigidity (price adjustment cost), which discourages the price setting behaviours of firms (Fernández-Villaverde et al., 2023)

IV. Concluding Remarks

In this paper, I study the ZLB issue generated by both supply and demand shocks using a stylized two-asset HANK model with real wage rigidity and an exogenous portfolio rule. ZLB leads to higher real bond return when nominal policy rate fails to respond to deflation in the economic. It necessarily makes bond repayment more costly for government and hence can impose tightening pressure on fiscal policy. Without funds from external sectors, fiscal policy will cyclically tighten at ZLB. The consumption-saving behaviours of wealth-poor households is more subject to the effect of this indirect fiscal response on their income level, while wealthy households are responsive to ZLB because the sizable interest income they earn dominates the effect of fiscal response. The results shows that poor households with high MPC are losers at ZLB and wealthy households with low MPC are winners, indicating less effectiveness of monetary policy in stimulating consumption and aggregate demand.

I prove that certain imperfections or micro-foundations in capital return or wage is required to generate responsive real marginal cost that would otherwise be fixed with perfectly flexible production cost paid for the provision of capital and labour. For instance, Kaplan et al. (2018) connects the monopolistic profit with capital return; Graves (2020) incorporates search and match frictions in the job market. I apply the reduced-form real wage rigidity to work around this property. Future work in HANK framework need to determine the appropriate micro-foundations in either return on private investment or labour market dynamics, depending on the relative importance of each factor for the research problem.

The reduced-form real wage rigidity I employ in the paper allows the real marginal cost and inflation to be computationally tractable. It adds a backward-dependent element to real marginal cost and inflation dynamics, allowing the rigidity to propagate to capital and bond returns and generating endogenous amplification of exogenous shocks in the model. Due to the backward dependent nature, the stronger the rigidity is in wage, the larger the deviation of real marginal cost from steady state and the more drastic the inflation dynamics is.

Consequently, the economy will be more prone to occurrence of ZLB events, and the duration will be prolonged.

There are two limitations of the model that prevent it to capture two propagation channels of ZLB. Firstly, aggregate employment is invariant to economic conditions because of the exogenous employment uncertainty. Secondly, the empirical portfolio rule restricts the extent to which households respond to endogenous investment environment via portfolio composition adjustment. Therefore, it is necessary to discuss how incorporation of the two channels can affect the results from the model.

In the TFP shock experiment, though ZLB leads to a decline of aggregate consumption, it is beneficial to the aggregate economy as households have more incentive to save in both capital and bond when bond return is higher at ZLB, narrowing the negative output gap. If households observe the shrinking spread between capital and bond return and further decline of capital return amplified by endogenously lower labour demand (wage is rigid downward and therefore employing labour is costly), it provides ample incentive for households to transfer their wealth into bond market, generating an endogenous 'crowding out' effect. For interest rate management policies to be effective in stimulating the economy, it would need to discourage households from invest in bond market and creating resource flows from bond holdings to both capital market and consumption, raising the aggregate demand and stabilize inflation.

In the positive preference shock experiment, aggregate demand increases despite of the decline of consumption. Though this is at odd with results in NK models, incorporating endogenous labour supply such that households are less willing to work in face of a positive preference shock can discourage households to invest in capital due to implied lower MPK, which can jointly produce downward pressure on aggregate demand. Change in aggregate demand would then become ambiguous and requires the determination of the net effect of increasing saving incentives from positive preference shocks and the impaired capital return due to less labour input. This is at least working towards reconciling the results of the current HANK model with standard narratives of positive preference shock in NK models. It is important to understand the net effect of a positive preference shock on aggregate demand because it determines the direction that policy makers should be exerting forces on. If aggregate demand increases, ZLB benefits the economy in closing the positive output gap by providing incentives to save in bond market. If aggregate demand falls short, expansionary monetary policies need to generate enough resource flows from bond market to capital market and consumption to recover aggregate demand as in a negative TFP shock.

A promising shock to the economy that would better capture a decline of aggregate demand is a direct disruption of the performance of capital market such as the GFC. An implicit assumption not commonly mentioned is that savings made by households can be perfectly translated into private investment and then perfectly translated into working capital without destruction (trial and error). The candidate shock can take the form of erosion rate in the process of translating savings into working capital. When it becomes riskier to invest in capital, it implies higher probability of investment erosion, households will find it more attractive to invest in bond instead of in capital. The insufficient demand of capital reduces private investment level and drives up capital return, reflecting the increase of erosion rate. Consumption will also decline because wealth level is impaired and higher perceived uncertainty and risk involved in capital investment leads to further precautionary saving in safe assets (government bond). Consequently, aggregate demand drops following the simultaneous decline in consumption and investment in capital – resource flows from capital and consumption to safe but unproductive assets, inducing deflationary pressure. This investment shock is different from TFP shock in the way that it does not have direct effect on the productivity of labour, but only through the change in working capital stock will labour productivity change and employment environment endogenous evolve.

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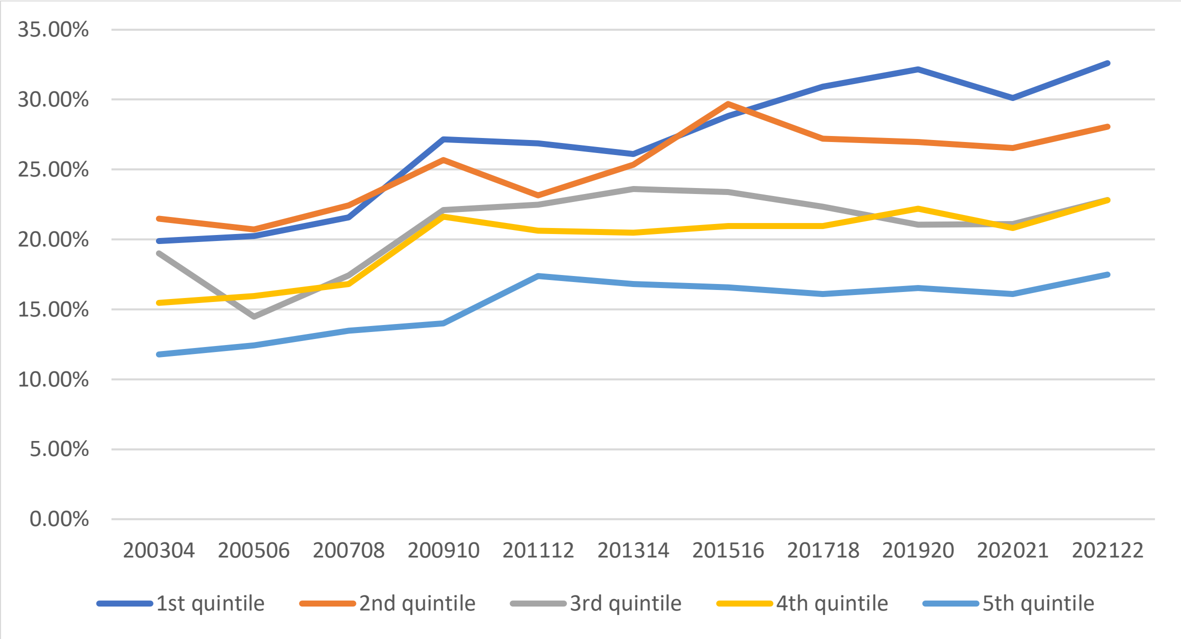
Appendix A. Empirical portfolio composition across wealth distribution



Appendix B. Algorithm and model solution

Appendix B.1. Steady state

1. Guess the steady state (K) and compute other state variables from the firm problem.
2. Solve the household problem and compute the new aggregate supply of capital and bond at the average of supply from households.
3. Compare the difference between the updated capital and the guessed capital.
   a. If the difference is larger than the exit criteria, update the guessed aggregate capital and recompute all the state variables and repeat Step 2.
   b. If the exit criteria is met, stop the iteration and compute the current aggregate and cross-sectional variables as their steady-state values.

Appendix B.2. Economic Response to Shocks

1. Specify the exogenous shock and guess the path of (K) and the paths of all the other variables using steady state values.
2. Iterate household problem backwards to update the path of the value functions and associated choice rules for consumption and asset holdings.
3. Simulate the impulse response functions of the economy forward and compare the difference between the path of aggregate capital supply from household problem and the guessed path of aggregate capital,

a. If the difference is larger than the exit criteria, update the path of aggregate capital and recompute the implied path of other state variables and return to Step 2.
b. If the exit criterium is met, stop looping and obtain the aggregate and household response to aggregate shocks.